SmartStudy Assistant

AI-powered adaptive learning and self-assessment

2nd Year, 2nd Semester

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Scan for the GitHub repository

1. Abstract

SmartStudy Assistant connects learning, assessment and revision in one web application. Students enter a topic or upload a text-based PDF. AI explains the material and creates questions. The application uses answer scores and concept tags to identify learning gaps, generates focused follow-up lessons, and schedules reviews with the SM-2 algorithm.

2. Introduction

Reading a lesson does not show whether a learner can explain or apply it. Students need feedback on specific mistakes and a practical way to revisit difficult concepts. This project combines these activities into a guided study cycle.

3. Problem Statement

Students may struggle to identify misunderstood concepts. Separate explanations, quizzes and revision tools make follow-up study difficult to organize. A connected workflow keeps learning gaps and future reviews in one place.

4. Objectives

1. Explain topics and PDF content.
2. Generate practice assessments.
3. Identify weak concepts.
4. Provide focused follow-up lessons.
5. Schedule future revision.
6. Retain study history and scores.

Acknowledgment

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5. System Components

Web interface
HTML, CSS and JavaScript

Express application
API routes, grading and study workflow

AI service
OpenRouter supplies model responses

PostgreSQL database
Sessions, questions, scores and reviews

6. Learning Workflow

1. Explain
Overview, examples and common mistakes.

2. Examine
MCQs and short-answer questions.

3. Diagnose
Scores and concept tags identify gaps.

4. Re-teach
Focused explanation and a mini-check.

5. Recall
Review due concepts and rate difficulty.

7. Assessment Logic

MCQs: compare the answer with the stored correct option.

Short answers: ask AI to evaluate understanding and return feedback.

Diagnosis: group incorrect or low-scoring answers by concept tag.

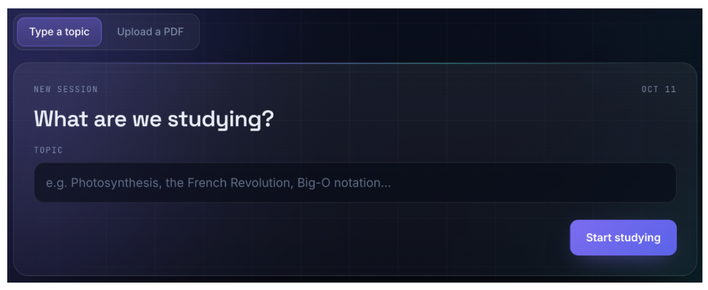
8. SM-2 Revision Scheduling

Initial review: high-severity gaps are due after one day; medium-severity gaps after two days.

Students rate recall as Again, Hard, Good or Easy. The SM-2 function updates the ease factor, repetitions and next review date.

Choosing Again resets the repetition count and schedules a review after one day.

9. Application Interface



Actual application interface: begin with a topic or upload a PDF.

10. Technologies and Data

Frontend: HTML, CSS, JavaScript.

Backend: Node.js and Express.

AI: OpenRouter via the OpenAI-compatible SDK.

PDF processing: multer and pdf-parse.

Storage: PostgreSQL / Supabase for sessions, assessments, answers, weak concepts and review schedules.

11. Discussion and Limitations

User IDs separate data, but password-based authentication is not implemented.

PDF input requires extractable text. OCR for scanned pages is not implemented.

AI explanations and grades can be incorrect.

Learning gains have not yet been established through a formal evaluation.

12. Conclusion and Future Work

The application connects explanation, testing and revision around each learner's weak concepts. Future work includes secure authentication, OCR for scanned PDFs, review notifications and evaluation with students.

Project Source

github.com/IMRAN-8/smartstudy-assistant

Thank you. Questions and feedback are welcome.